

Abstract No.

CBLs - Towards a skull-based T-cell therapy for glioblastoma treatment.

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Glioblastoma is among the most pressing unmet medical needs in oncology. Despite maximal safe tumor resection and temozolomide-based chemoradiotherapy, the median overall survival remains poor with approximately 18 months.

We recently identified human skull marrow as a previously unrecognized reservoir of tumor-reactive T cells in patients with glioblastoma. CD45+ immune cells recovered from this compartment were predominantly T cells, which we termed Cranial Bone Lymphoids (CBLs). Integrated single cell gene transcriptomic and T-cell receptor profiling revealed frequent tumor-reactive clonotypes and substantial clonal overlap with tumor-infiltrating T-cells. CBLs exhibited a phenotype consistent with preserved functional fitness and sustained repeated ex vivo expansion capacity, supporting their potential for autologous adoptive T-cell therapy.

Here, we report the initial translational steps toward developing CBLs as an autologous advanced therapeutic medicinal product (ATMP) for glioblastoma. We compare workflows for cell isolation and processing, determine ex vivo expansion capacity, and characterize the cellular composition of the resulting products. We further assess CBL-mediated cytotoxic against autologous tumor cells to evaluate its suitability as a product-specific potency readout. These studies address key feasibility parameters for subsequent process standardization and GMP-compliant manufacturing development.

This work provides the basis for advancing CBLs towards a first-in-human, point-of-care cellular immunotherapy for glioblastoma.



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